



Introducing Cloud Architecting

AWS Academy Cloud Architecting



Introduction

Introducing Cloud Architecting

Module objectives

This module prepares you to do the following:

- Define cloud architecture.
- Describe how to design and evaluate architectures using the AWS Well-Architected Framework.
- Explain best practices for building solutions on Amazon Web Services (AWS).
- Describe how to make informed decisions about where to place AWS resources.

Module overview

Presentation sections

- Cloud Architecting
- AWS Well-Architected Framework
- Best practices for building solutions on AWS
- AWS Global Infrastructure

Knowledge checks

- 10-question knowledge check

As a cloud architect designing AWS architectures:



- I need to understand the practice of applying cloud best practices to a solution that uses cloud services and features to meet an organization's technical needs and business use cases.
- I need to evaluate an architecture with the AWS Well-Architected Framework.
- I need to apply best practices to building solutions on AWS.



Cloud Architecting

Introducing Cloud Architecting

Cloud computing and AWS

2000



Amazon was challenged with building an ecommerce service to for third-party sellers to build their own shopping sites.

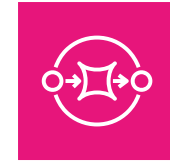
Early 2000s



Amazon developed well-documented APIs.

Projects still took months to complete.

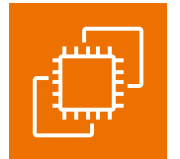
2006



Amazon SQS



Amazon S3



Amazon EC2

Cloud architecture



Requirements



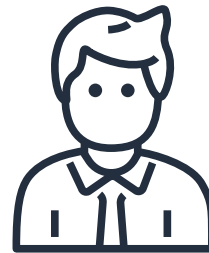
Structure design



Completed structure



Customer
(Decision maker)



Architect



Building crew
(Delivery team)

Role of a cloud architect



Plan

- Set technical cloud strategy with business leads.
- Analyze solutions for business needs and requirements.



Research

- Investigate cloud services specifications and workload requirements.
- Review existing workload architectures.
- Design prototype solutions.



Build

- Design the transformation roadmap with milestones, work streams, and owners.
- Manage the adoption and migration.

Key takeaways: What is cloud architecting?



- Cloud architecture is the practice of applying cloud characteristics to a solution that uses cloud services and features to meet an organization's technical needs and business use cases.
- You can use AWS services to create highly available, scalable, and reliable architectures.
- Cloud architects are responsible for managing an organization's cloud computing architecture.



AWS Well-Architected Framework

Introducing Cloud Architecture

Pillars of the AWS Well-Architected Framework



Operational
Excellence



Security



Reliability



Performance
Efficiency



Cost
Optimization

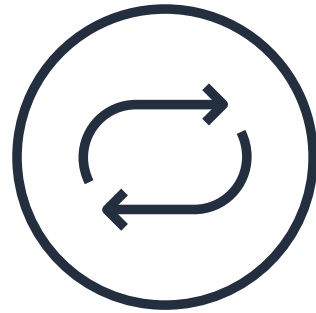


Sustainability

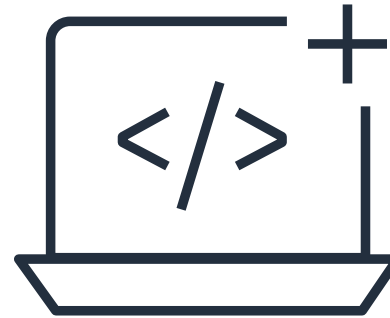
Operational Excellence pillar



Run and monitor systems that deliver business value.



Continually improve supporting processes and procedures.



View the entire workload as code.

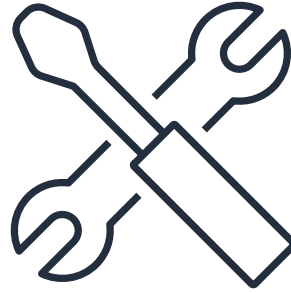


Operational
Excellence

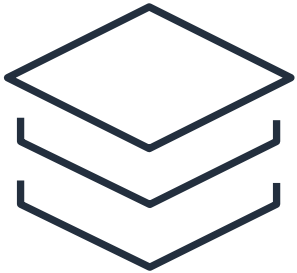
Security pillar



Implement a strong identity foundation.



Maintain traceability.



Apply security at all layers.

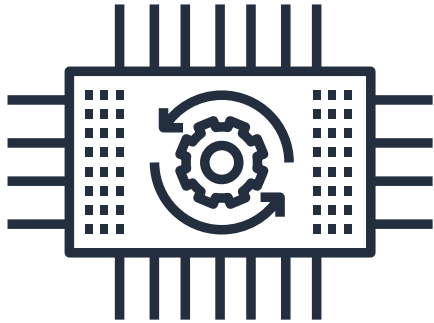


Implement risk assessment and mitigation strategies.



Security

Reliability pillar



Recover quickly.



Dynamically meet
compute demand.



Mitigate disruptions.



Reliability

Performance Efficiency pillar



Choose and maintain efficient resources.



Democratize advanced technologies.



Employ mechanical Sympathy.



Performance Efficiency

Cost Optimization pillar



Measure efficiency.



Adopt the right
consumption model.



Eliminate unneeded
expense.



Consider using
managed services.



Cost
Optimization

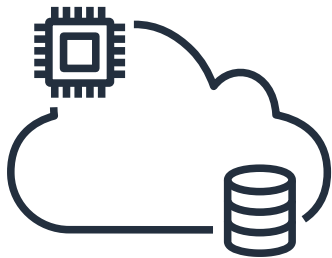
Sustainability pillar



Establish sustainability goals.



Maximize utilization.



Choose efficient hardware
and software.



Reduce downstream impact.



Sustainability

Using the AWS WA Tool



AWS Well-Architected Tool

- Helps you review the state of your workloads and compares them to the latest AWS architectural best practices
- Gives you access to knowledge and best practices used by AWS architects when you need it
- Delivers an action plan with step-by-step guidance on how to build better workloads for the cloud
- Provides a consistent process for you to review and measure your cloud architectures

Key takeaways: AWS Well-Architected Framework



- The AWS Well-Architected Framework provides a consistent approach to evaluate cloud architectures and guidance to help implement designs.
- The AWS Well-Architected Framework is organized into six pillars.
- Each pillar documents a set of foundational questions you can use to understand if a specific architecture aligns well with cloud best practices.
- The AWS WA Tool helps you review the state of your workloads and compares them to the latest AWS architectural best practices.



Best practices for building solutions on AWS

Introducing Cloud Architecting

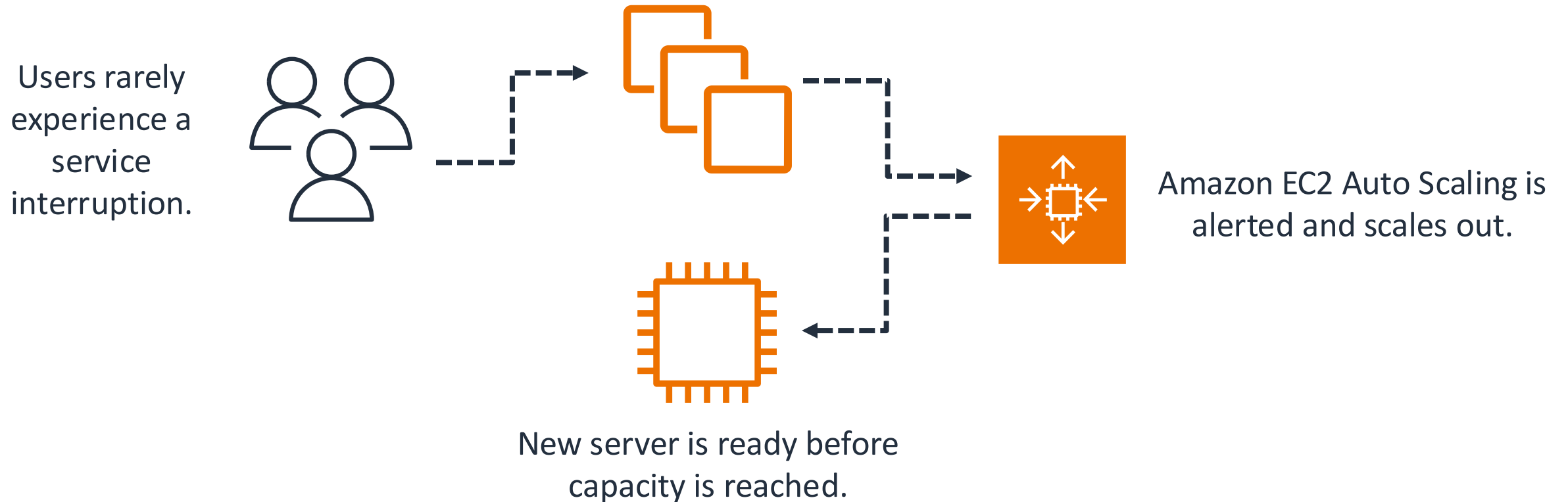
Design trade-offs

As you design a solution, think carefully about trade-offs so you can select an optimal approach.

- Evaluate trade-offs so you can select an optimal approach.
- Examples of trade-offs include the following:
 - Trade consistency, durability, and space for time and latency to deliver higher performance.
 - For new features, prioritize speed to market over cost.
- Base design decisions on empirical data.

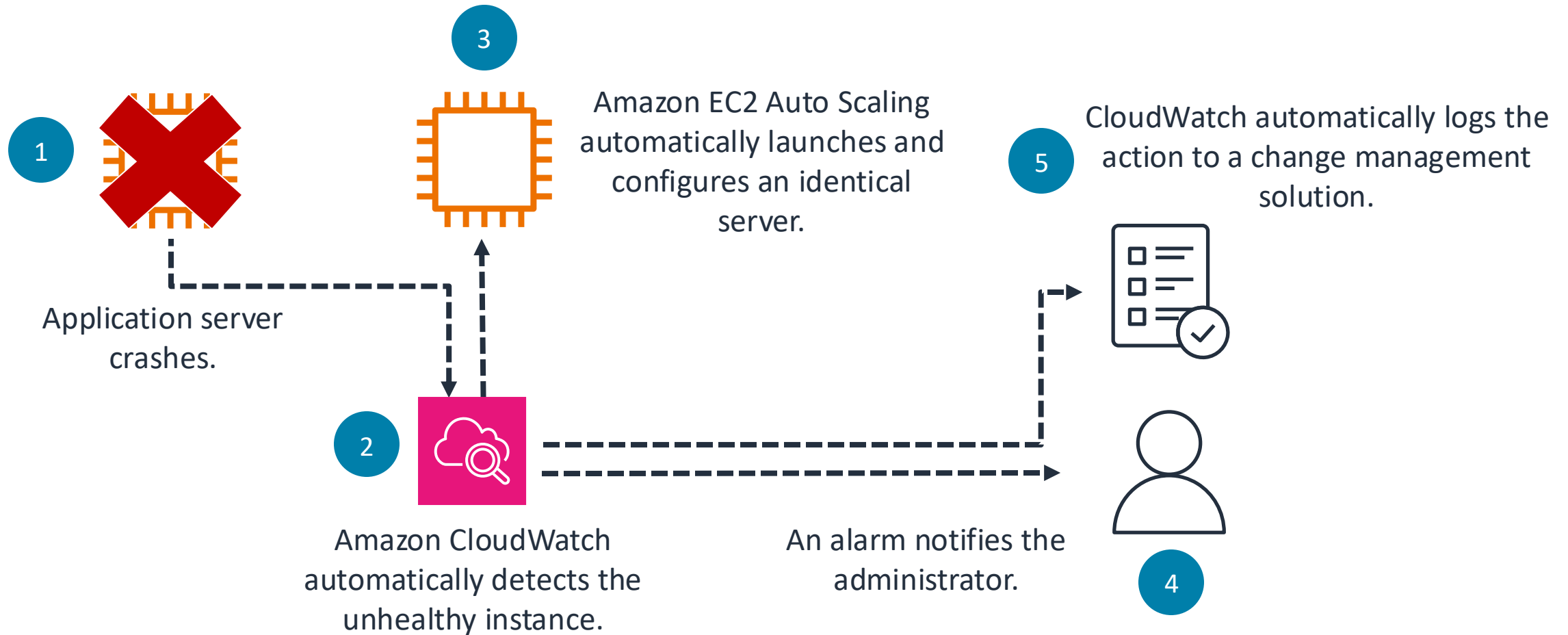
Implementing scalability

Ensure that your architecture can handle changes in demand.



Automating your environment

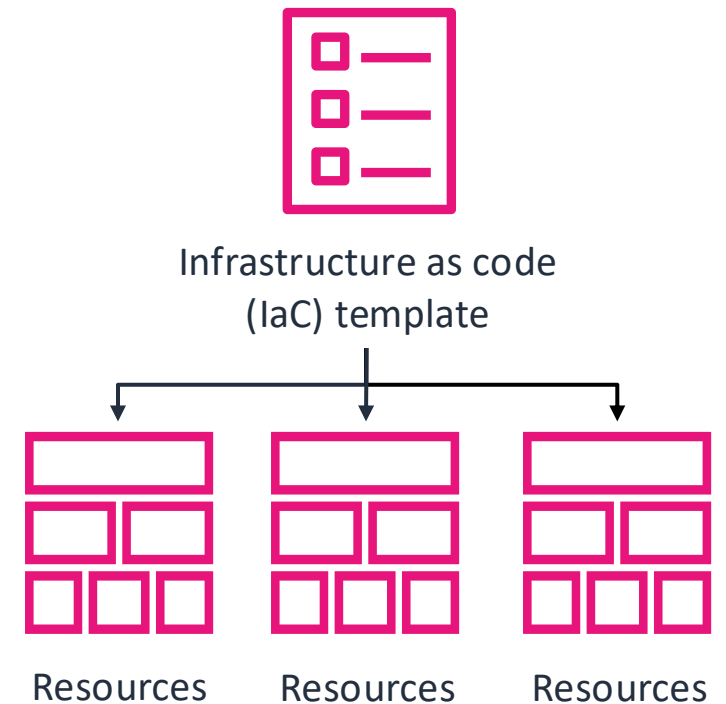
Automate the provisioning, termination, and configuration of resources.



Using IaC

Provision your computing infrastructure using code instead of manual processes.

- Rapidly deploy duplicate environments.
- Reduce configuration errors from manual configuration.
- Propagate changes consistently to all stacks.



Treating resources as disposable

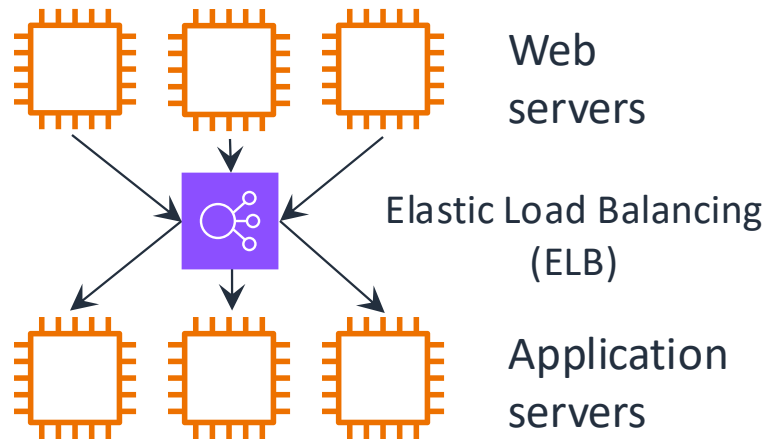
Take advantage of the dynamically provisioned nature of cloud computing.

- Automate deployment of new resources with identical configurations.
- Stop resources that are not in use.
- Test updates on new resources, and then replace old resources with updated ones.

Using loosely coupled components

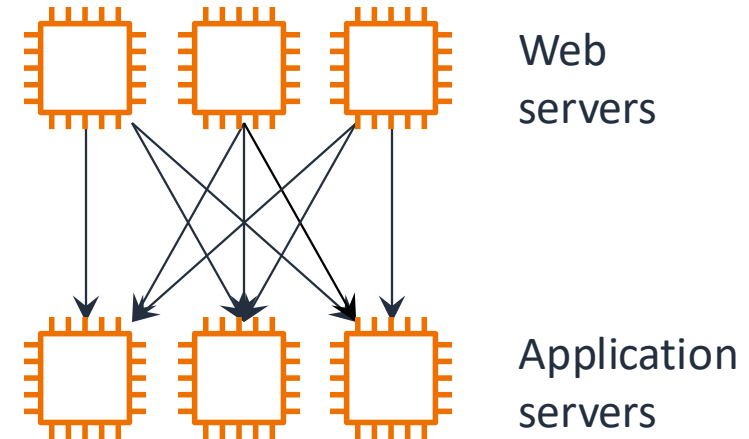
Design architectures with independent components.

Best practice



Web servers decoupled with Elastic Load Balancing (ELB).

Anti-pattern



Web servers tightly coupled to application servers.

Designing services, not servers

Use the breadth of AWS services. Don't limit your infrastructure to servers.

- When appropriate, consider using containers or a serverless solution.
- Message queues can handle communication between applications.
- Static web assets can be stored off server, such as on Amazon Simple Storage Service (Amazon S3).
- User authentication and user state storage can be handled by managed AWS services.

Choosing the right database solution

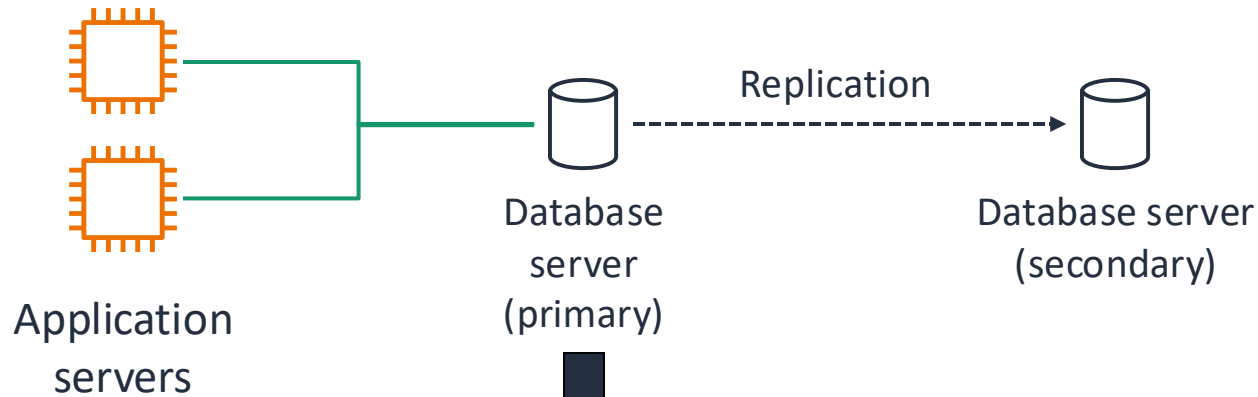
Match technology to the workload, not the other way around.

- Read and write needs
- Total storage requirements
- Typical object size and nature of access to these objects
- Durability requirements
- Latency requirements
- Maximum concurrent users to support
- Nature of queries
- Required strength of integrity controls

Avoiding single points of failure

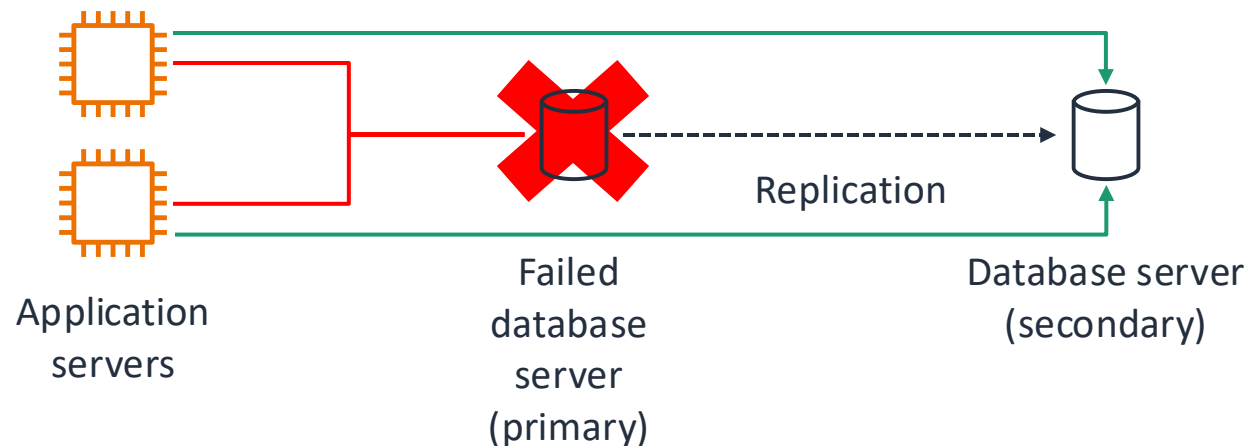
Assume everything fails. Then, design backward.

Under normal operations



Create a secondary (standby) database server and replicate the data.

Upon failure



If the main database server goes offline, the secondary server picks up the load.

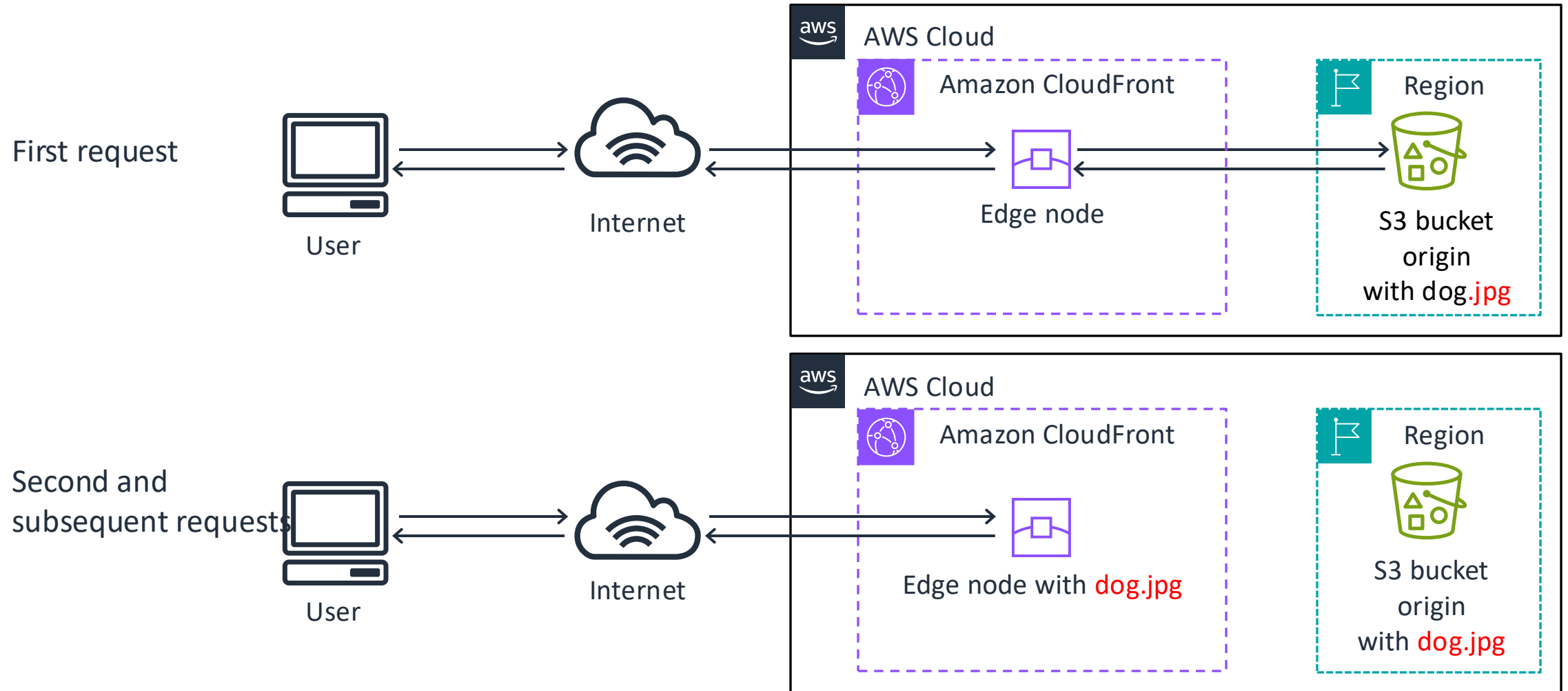
Optimizing for cost

Take advantage of the flexibility of AWS to increase your cost efficiency.

- Are my resources the right size and type for the job?
- Which metrics should I monitor?
- How do I turn off resources that are not in use?
- How often will I need to use this resource?
- Can I replace any of my servers with managed services?

Using caching

Minimize redundant data retrieval operations, improving performance and cost.



Securing your entire infrastructure

Build security into every layer of your infrastructure.

- Use managed services.
- Log access of resources.
- Isolate parts of your infrastructure.
- Encrypt data in transit and at rest.
- Enforce access control granularly, using the principle of least privilege.
- Use multi-factor authentication (MFA).
- Automate your deployments to keep security consistent.

Key takeaways: Best practices for building solutions on AWS



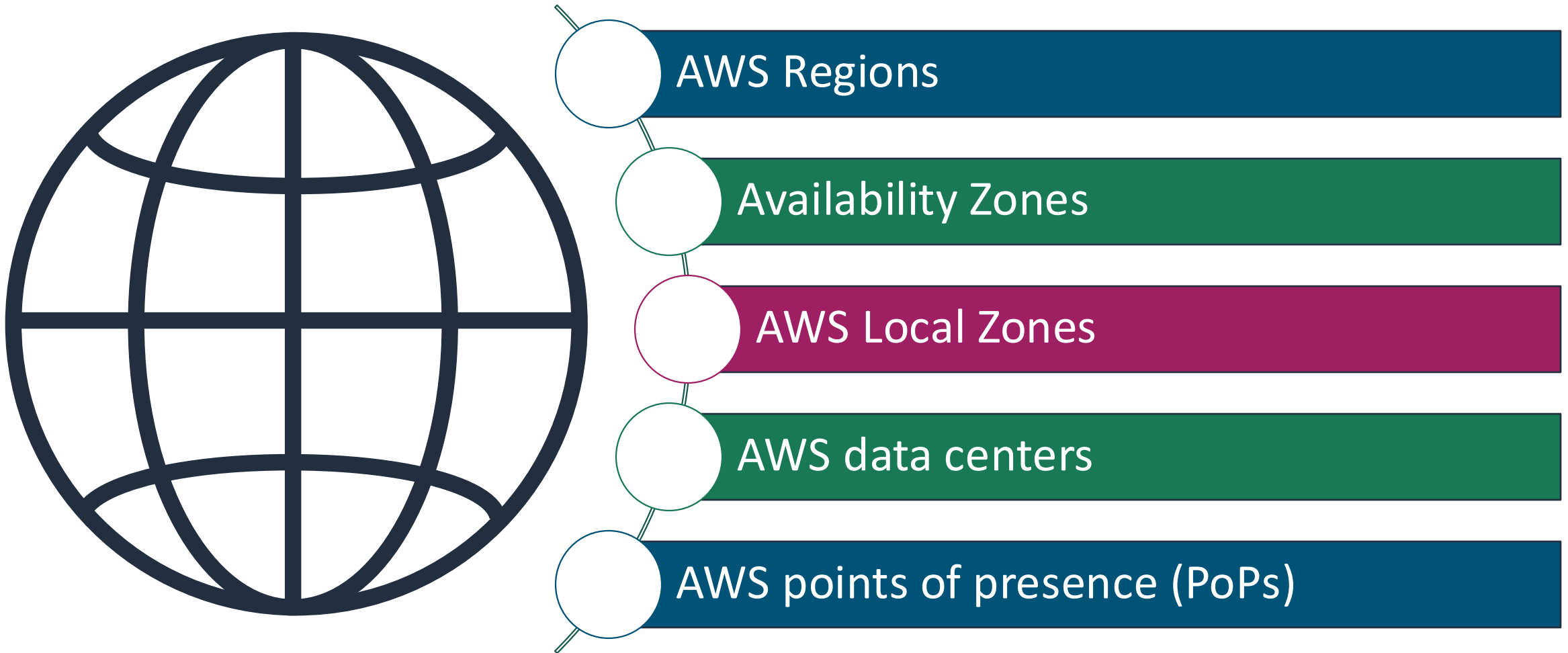
- As you design solutions, evaluate trade-offs and base your decisions on empirical data.
- Follow these best practices when building solutions on AWS:
 - Implement scalability.
 - Automate your environment.
 - Treat resources as disposable.
 - Use loosely-coupled components.
 - Design services, not servers.
 - Choose the right database solution.
 - Avoid single points of failure.
 - Optimize for cost.
 - Use caching.
 - Secure your entire infrastructure.



AWS Global Infrastructure

Introducing Cloud Architecting

AWS infrastructure topics



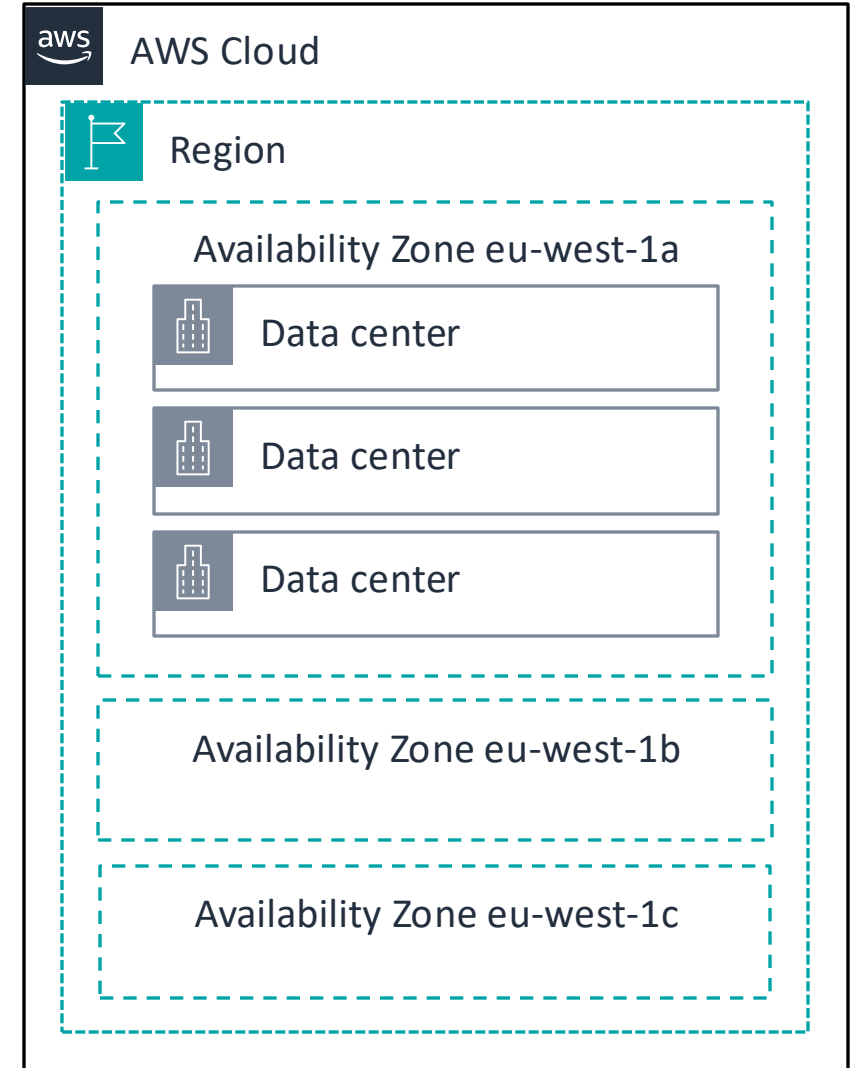
Selecting Regions

- A Region is a geographical area.
- Each Region usually consists of two or more Availability Zones.
- Communication between Regions uses AWS backbone network infrastructure.
- You enable and control data replication across Regions.



Selecting Availability Zones

- Each Availability Zone includes the following:
 - It is made up of one or more data centers.
 - It is designed for fault isolation.
 - It is interconnected with other Availability Zones in a Region using high-speed private links.
- For certain services, you can choose your Availability Zones.
- AWS recommends replicating across Availability Zones for resiliency.



Using Local Zones

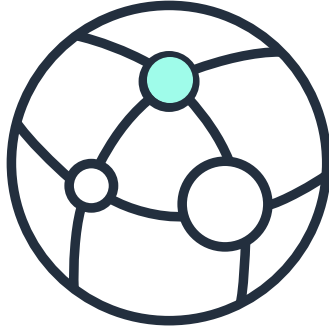
- Local Zones make it possible for you to run latency-sensitive portions of applications closer to end users and resources in a specific geography.
- They are an extension of a Region.
- With Local Zones, you can place AWS compute, storage, database, and other select services closer to large population, industry, and IT centers where no Regions exist today.
- Local Zones are managed and supported by AWS.

Role of AWS data centers

- Data centers are where the data resides and data processing occurs.
- A data center typically has tens of thousands of servers.
- All data centers are online and serving customers.
- AWS custom network equipment includes the following:
 - Is sourced from multiple ODMs
 - Has a customized network protocol stack

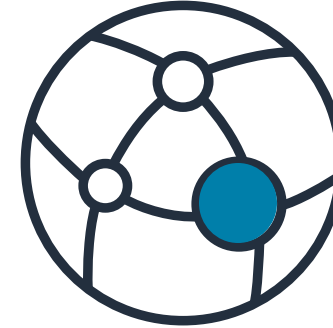


AWS PoPs



Edge location

AWS data centers and servers located close to customers and designed to deliver services with the lowest latency possible



Regional edge cache

AWS data centers between the origin server and the edge location that have a longer cache

Key takeaways: AWS Global Infrastructure



- The AWS Global Infrastructure consists of Regions, Availability Zones, and edge locations.
- Your choice of a Region is typically based on compliance requirements or to reduce latency.
- Each Availability Zone is physically separate from other Availability Zones and has redundant power, networking, and connectivity.
- Edge locations and regional edge caches improve performance by caching content closer to users.



Module wrap-up

Introducing Cloud Architecting

Module summary

This module prepared you to do the following:

- Define cloud architecture.
- Describe how to design and evaluate architectures using the AWS Well-Architected Framework.
- Explain best practices for building solutions on AWS.
- Describe how to make informed decisions about where to place AWS resources.

Module knowledge check



- The knowledge check is delivered online in your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.



Thank you

Corrections, feedback, or other questions?

Contact us at <https://support.aws.amazon.com/#/contacts/aws-academy>.